Overview of Frequency Control Techniques in Power Systems with High Inverter based Resources: Challenges and Mitigation Measures

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Abstract: Power systems are rapidly transitioning toward having an increasing proportion of electricity from inverter-based resources such as wind and solar. An inevitable consequence of a power system transition towards 100% inverter-based resources is the loss of synchronous generators with their associated inertia, frequency, and voltage control mechanisms. To ensure future grid needs can be met, the source of system services to meet these needs must change from synchronous plants to inverter-based resources. Under this context, the main objective of this research is to extensively review grid frequency stability challenges concerning the massive integration of inverter-based resources from the perspective of system operators. Following that, the newly established international fast frequency response services in different renewable dominant power systems to address low inertia challenges are compared from various perspectives. Finally, potential solutions based on various renewable enabling technologies and renewable sources are investigated from the standpoint of their expectation to play an active role in the energy system and to provide a wide range of system frequency services. The findings of this study provide a deep understanding of prospective frequency control solutions and an actionable dataset that will support various research and policy needs as we approach net-zero targets.

Keywords: Distribution generation, Grid forming converter, Grid following converter, Low inertia power systems, Renewable energy, Virtual synchronous machine

1. Introduction

The energy sector must be fundamentally restructured globally to ensure complete decarbonization. One of the consequences was the establishment of the Paris Agreement in December 2015, in which participants committed to take the necessary actions to limit global rising temperature to less than 2 degrees Celsius (°C), with attempts to limit it to below 1.5 °C [1]. Achieving such ambitious goals requires full decarbonization of the energy sector. The outcomes of the 2018 assessment of the Intergovernmental Panel on Climate Change (IPCC) were clear: to have any chance of enduring the Paris Agreement and minimizing the potential impacts of catastrophic runaway global warming, the world must be carbon-neutral by 2055 [2]. In all its forms, renewable energy has been considered as one of the essential drivers to meeting the objectives of the Paris Agreement. It can be argued that electrical power systems are among the easiest energy supplies to decarbonize compared to transportation and heating. Fig. 1 depicts global installed renewable electricity, including renewable generation and its share of overall power generation [3], [4]. In 2020, these sectors were 2802 GW, 7643 TWh, and 28.6%, respectively.

The Paris Agreement anticipates that the proportion of worldwide annual renewable electricity generation will increase from about 28% today to 86% in 2050 [5]. Distributed generation (DG) based on variable renewable (i.e., wind and solar photovoltaic (PV) are projected to account for about 70% and about 60% of worldwide annual electricity generation [5]. Regional efforts are also worth mentioning. For instance, as part of the climate action plan, the European (EU) countries have set ambitious targets to produce 54% of the gross generated electricity mix from renewable sources by 2030 [6] and reach net zero by 2050. Distributed generation-based wind power and PV systems are key drivers toward these objectives accounting for 56% of Europe's electricity by 2030 and up from

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27% in 2021 [7]. Among the EU countries, renewable generation in Ireland accounted for 43% of the overall electricity consumption in 2020 [8]. Ireland has recently released the Climate Action Plan 2021 (CAP21), which sets itself the challenge of sourcing 70%, which may increase to 80%, of its electricity generation from renewable energy, predominantly through variable speed wind turbines, by 2030 [9]. The CAP21 also contains a higher commitment to deploying up to 5 GW of offshore wind installed capacity. In the United Kingdom (UK), the government is aiming toward net zero carbon emissions by 2050, but no precise target has been set for Northern Ireland, which may reach 80% by 2030 [10].

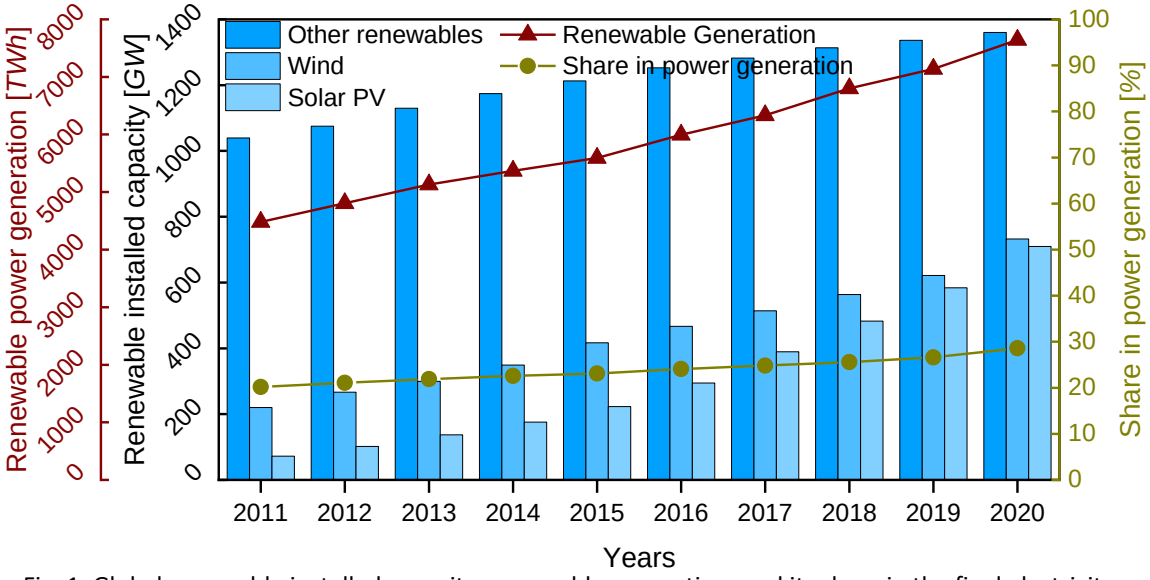


Fig. 1. Global renewable installed capacity, renewable generation, and its share in the final electricity generation.

This means that power systems are moving away from large, centralized generating units toward high levels of variable DG paradigm, in which renewable sources (i.e., wind and solar) will play a key role. This shift in generation technology changes the fundamental technical characteristics of the network. This new form of power generation poses grid instability and vulnerability challenges due to the intermittency and nonsynchronous nature of renewable energy if operated in grid following mode (GFL) [11]. Currently, uncertainties associated with the operation of variable DG systems are handled utilizing the spinning reserve of conventional generators [12]. However, traditional power systems were not built to handle growing uncertainty; thus, significant penetrations of DG systems necessitate some predictability to prevent the possibility of a mass blackout.

An equally essential challenge is managing grid reliability with a high level of nonsynchronous nature of DG systems (i.e., wind and solar PV), as well as high voltage direct current (HVDC) interconnectors. These are mostly inverter-based resources (IBR) that are decoupled from the utility grid via power electronic converters and are referred to in this study as those that operate in GFL mode. Consequently, the inertia of their rotating mass, as in wind turbines, is not exposed to the transmission network if it is not operated in a frequency-sensitive mode [13]. Large penetrations of IBR at the expense of synchronous machines further displace the amount of spinning mass and result in a significant reduction in the overall power system inertia [14]. In other words, the inertia makes the system reluctant to a change in angular velocity and frequency [15]. This may require renewable generations to be curtailed, forcing transmission system operators (TSO) to rely on more fossil fuel-based plants until adequate inertia and reserve are available to manage contingency events and keep the system running at a nominal frequency.

Aside from synchronous inertia response (SIR), synchronous generators with turbine governors provide another essential service in the form of a droop, or governor response, following an

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unplanned generation/load imbalance. When a large number of synchronous generators are replaced by grid-scale variable renewable sources operating at maximum power point tracking (MPPT), the system loses both inertia and primary operating reserve (POR) capabilities. This means that the grid is less able to handle the loss of large infeeds, and is not as effective at maintaining stability during severe contingencies [12]. Thus, the instantaneous balance between generation, demand, and power grid management in real-time is becoming a challenging task for TSOs [16]. A direct threat from low system inertia and spinning reserve is a larger and faster rate of change of frequency (RoCoF) with higher frequency deviation (nadir) [17]. Extreme RoCoF and frequency nadir are highly undesirable. They contribute to triggering the anti-islanding DG protection relays followed by load shedding, leading to cascading failures and eventually complete system blackout similar to that which occurred in the South Australia 2016 blackout [18]. To address these issues, a number of islanded countries, including Ireland, the UK, and Australia, have recently established a fairly new fast frequency response (FFR) ancillary services market [19]. The FFR service is designed to help TSOs maintain a stable operating frequency, thereby facilitating the integration of high IBR [20].

In addition to transmission system changes, the electrification of end-use sectors such as transportation (i.e., electric vehicles (EV)), building (i.e., heat pumps (HP), and industry (i.e., data centers) is radically reshaping the distribution network. This is occurring in tandem with the decarbonization of the power sector. In this context, there is now a higher need for system flexibility to deal with both transmission and distribution system challenges. This flexibility will be used for various reasons and will be derived from a variety of sources throughout the power value chain, from generation to demand. International Renewable Energy Agency (IRENA) defines flexibility as the capability of a power system to cope with the variability and uncertainty that solar and wind energy introduces at different time scales while minimizing renewable power curtailment and reliably delivering energy demand to all customers [21]. However, reaching this goal will require load centers on the grid to be more flexible for three reasons, (1) better match the grid's increasingly unpredictable supply, (2) fine-tune the grid for transient stability, (3) and control power flows on the network architecture. These requirements will not only become more pronounced in power systems with high IBRs, but the conventional fossil-fuel generators that will be employed to address these flexibility needs will also become more expensive to operate. As a result, other resources (i.e., synchronous condensers, energy storage, and demand response schemes) will be required to provide power system frequency needs through participating in the ancillary services [22], [23].

The majority of the aforementioned issues, including the effect of IBR on power system frequency stability [24], inertia level [25], and voltage control mechanisms [26], have been addressed in recent research. However, none of them evaluated the challenges and potential technology solutions in the context of system operators' plans to achieve net zero targets. Different from the literature, the main objective of this study is to comprehensively review strategies used to improve frequency control and manage RoCoF in both islanded and interconnected power systems. The scope of the analysis then extends to frequency support mechanisms from wind and solar systems, HVDC interconnectors, synchronous and asynchronous condensers, battery energy storage systems (BESS) equipped with both grid forming (GFM) and GFL converters.

The remainder of the paper is structured as follows. Section 2 examines the impact of IBR on frequency stability and RoCoF management in weak and well-interconnected power systems. In Section 3 deep understanding of international frequency service and frequency response enabling technologies are critically evaluated. Discussions and recommendations are provided in Section 4. Section 5 concludes the paper with some remarkable future work directions.

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2. Impacts of inverter-based resources on frequency stability

The state-of-the-art DG-based renewable generation technologies (i.e., wind and solar PV plants) are integrated into the grid through power converters. The massive integration of these DG systems into the transmission and distribution networks impacts the overall dynamic stability of the power system [27]. This is primarily caused by two main changes in the network topology: displacement of conventional power plants and rapid time constants introduced by power electronic converters. Furthermore, power systems are shifting from centralized massive synchronous generators to decentralized generation at the distribution level [28]. The latter poses a two-fold challenge: 1) observing the generation fleet in medium and low voltage network infrastructure becomes difficult for TSOs, and 2) load flow patterns in transmission and distribution networks become volatile, leading to greater reverse flows and faster voltage instability across network nodes [29]. All these changes introduce several stability issues that are commonly classified, as Fig. 2 illustrates [30].

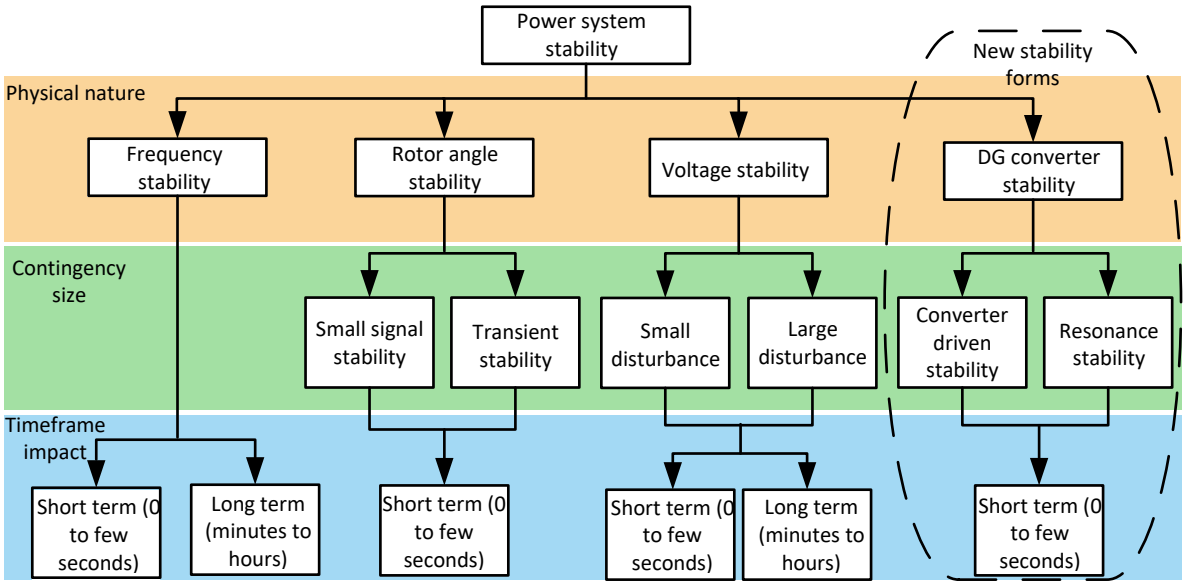


Fig. 2. Classification of power system stability issues.

With the high penetration of IBRs, two new components of power system stability have recently emerged: (1) converter-driven stability (oscillations due to cross-coupling between dynamic electromechanical devices and network transient), and (2) resonance stability (oscillations due to periodic and inadequate system energy dissipation) [30]. Finally, frequency stability is explained in the following sections, while voltage and rotor angle stabilities are out of the scope of this paper.

Grid frequency is an important aspect of a power system that enables paralleled conventional plants to operate cooperatively. Regardless of the original power sources, all generators tend to react the same way; if demand exceeds generation, the system frequency drops, with a higher degree of power imbalances resulting in larger frequency deviations [31]. Implying that the system frequency is a key metric to define a balance between generation and demand and should be kept within a permissible range. Under frequency is a challenge to deal with, especially during high intermittent renewable penetrations, since additional power must be injected into the grid to halt the decline and thus is subject to the availability and source of power reserve. However, over frequency is usually easier to manage because generation technology can swiftly decrease output power and rebalance supply and demand [32].

2.1. Frequency response of conventional power systems

Frequency deviations are frequently caused by disturbances, which must be addressed by several ancillary system services at the transmission level. Traditionally, the primary operating reserve (POR)

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of synchronous generators is utilized to maintain frequency deviations during the arresting period. Then the secondary and tertiary responses are activated to recover frequency back to a quasi-steady state and resume an economically efficient system dispatch [33]. If the primary response fails to contain a frequency drop, the under frequency load shedding (UFLS) may be triggered, potentially resulting in further cascade issues similar to the disruption seen in the United Kingdom (UK) in 2019 [34]. To protect against such circumstances, the TSOs of Ireland and Northern Ireland established a long-term comprehensive work plan, namely Delivering a Secure, Sustainable Electricity System (DS3) in 2011 [35]. The main objective is to tackle power quality issues and to provide the conditions required to securely operate at system nonsynchronous penetration (SNSP) up to 75%. Both TSOs have made significant progress in this approach by designing and implementing new frequency control services that address a wide range of applications (i.e., response, reserves, and ramping capabilities) and timelines from milliseconds to hours to complement the existing services.

An example of the DS3 system services in the Irish power system and detailed control responses that are used during each time scale is depicted in Fig. 3. The graph illustrates a typical frequency response following a relatively large generation loss that results in frequency deviation from steady-state value. The generation and demand are balanced at $t = 0^-$, and the system frequency is at a nominal value. At $t = 0^+$, there is an unanticipated loss of generation or demand increase resulting in a quick frequency decline. The time window that follows a contingent event and ends at the minimum frequency (nadir) is referred to as the arresting phase. The SIR and FFR services, in particular, have been established for providing an effective response within the arresting period. Initially, the stored energy in the rotating masses of all grids connected synchronous machines delivers in the form of SIR during the arresting phase. While SIR is beneficial in slowing the rate of change of frequency (RoCoF), if no action is taken, the load and generation imbalance will continue to diverge system frequency from the nominal value. This means that SIR alone is not enough to stop frequency deviation following the disturbance; rather, SIR buys time for other means to act within this phase.

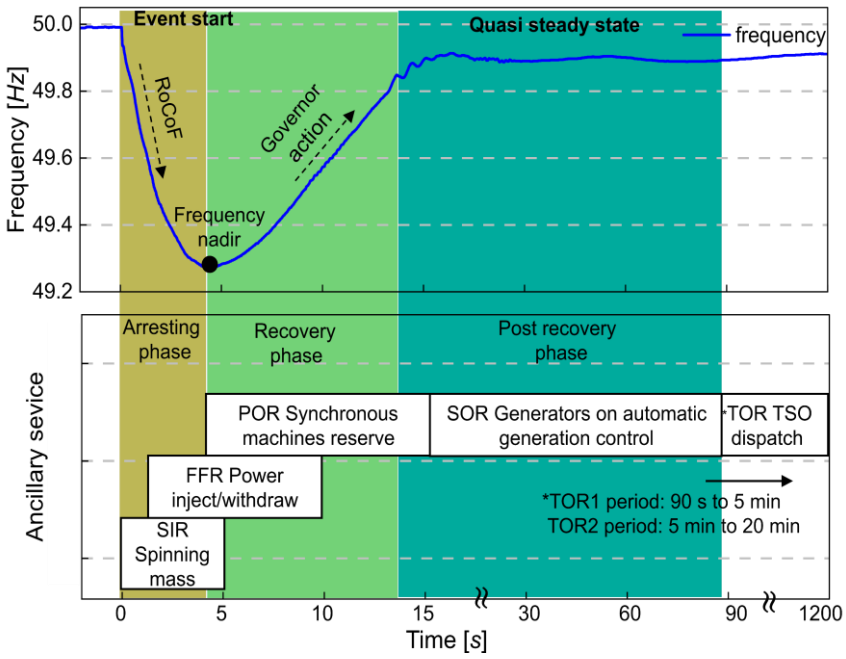


Fig. 3. Timescale of frequency response services in the Irish power sustem following a typical disturbance.

Following the SIR service, traditional thermal power plants with turbine governors begin responding to system frequency during this phase, providing POR to arrest frequency deviation. The POR is a contingency reserve responsible for arresting the initial drop and restoring the frequency, which

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occurs at the level of individual controllers via droop control mechanisms. At frequency nadir, a new rebalance between generation and demand typically occurs. Various factors affect the severity of nadir and none of them can be examined in isolation without the other being considered. These factors include the overall system inertia, disturbance size, response speed of POR (i.e., gas or hydro turbines), the magnitude of response, and load sensitivity to frequency deviation [36]. A combined response of POR and secondary operating reserve (SOR) (i.e., automatic generation control) in the timescale 5 – 15 s and 15 – 90 s, respectively, eventually recover frequency to a post-disturbance value.

Many of these services are provided by traditional thermal power plants. As these plants become less common in the future, reducing the overall system inertia and resulting in more frequent high RoCoF values during disturbances [37]. This means that the remaining synchronous generators will have less time to act during the arresting phase. Thus, both TSOs on the island of Ireland have established the complementary FFR service, which relies on alternative sources to provide a faster form of system reserve to bridge the gap between SIR and POR services. The FFR service is the most lucrative, requiring active power to be injected in less than 2 s from the time of the disturbance in response to both over and under frequency events [38]. As the main objective of the FFR service is to arrest frequency excursion quickly before reaching extreme nadir, it overlapped with both SIR and POR services. This means the amount and quality of POR supplied reveal how much FFR is required and how effective it is [39]. Faster POR, for example, will minimize the required FFR capacity at any given level of inertia.

2.2. Power system inertia

Inertia is an inherent characteristic of spinning machines in a power system that resists changes in rotational speed [40]. It is a physical behavior of the system that plays an essential role in limiting RoCoF during disturbances. Inertia is available immediately and is typically provided by synchronous generators, synchronous condensers, and some spinning demand loads. It is a key determinant of the strength and stability of the power system. The amount of inertial response in a system is proportional to the total number of synchronous generators, turbines, and the speed of rotation of each rotating mass, which can be determined using (1).

$$H_{sys} = \frac{E_k}{S_{sys}} = \frac{1}{2}\frac{\omega^2}{S_{sys}} \sum_{i=1}^{n} J_i \qquad \forall\, i \in R \qquad (1)$$

where H_{sys} is the system inertia constant in (s), E_k denotes the total kinetic energy in (Joule), S_{sys} is the system base in (MVA), ω presents the rotor rotational speed in (rad/s), and J_i moment of inertia of synchronous machine i in (kg.m^2), n is the number of online spinning machines. From (1), the complete system inertia constant H_{sys} containing n numbers of machines can be reformulated using (2).

$$H_{sys} = \sum_{i=1}^{n} H_i . S_i / S_{sys} \qquad (2)$$

where H_i inertia constant of machine i, S_i nominal power of machine i. The level of H_{sys} is an indicator that defines the degree and quantifies robustness of system frequency in interconnected networks. The RoCoF in the frequency arresting phase is a direct metric that reflects the amount of available H_{sys} as in (3).

$$\frac{df}{dt}\Big|_{t=0^+} = \frac{f_o}{2H_{sys}S_{sys}}\left(P_g - P_l\right) = \frac{\Delta P_{imbalance}\, f_o}{2H_{sys}S_B} \qquad (3)$$

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where df/dt is the RoCoF in (Hz/s), $P_g - P_l = \Delta P_{imbalance}$ depicts the generation and demand imbalance in (MW), and f_o is the nominal frequency of the system in (Hz). Extreme RoCoF values are clearly caused by a higher power loss or lower system inertia. The amount of H_{sys} decreases as more conventional synchronous generators are replaced by GFL-based IBRs. As seen in Fig. 4, for the same disturbance (i.e., infeed loss), increasing the penetration of GFL-based IBR in the system results in extreme frequency deviation and RoCoF [16]. This is an example of a power system in which governor-equipped synchronous generators with their associated inertia are replaced with GFL-based IBRs operating at MPPT while the system still has a large amount of slow secondary reserve. This graph indicates that by lowering inertia, the power system dynamics become considerably faster, requiring faster measurement, which necessitates less latency in RoCoF measurement techniques. This raises the risk of maximum RoCoF detection inaccuracy in the future [41] and a challenge of regional RoCoF variations [42]. This is because the maximum RoCoF occurs within the first second following the disturbance, and when the frequency settles at the nadir, the RoCoF becomes zero. Furthermore, a high RoCoF is typically problematic since it shortens the time window for POR service to control the frequency deviation. Physically, DG and conventional power plants are not designed to tolerate high RoCoF events.

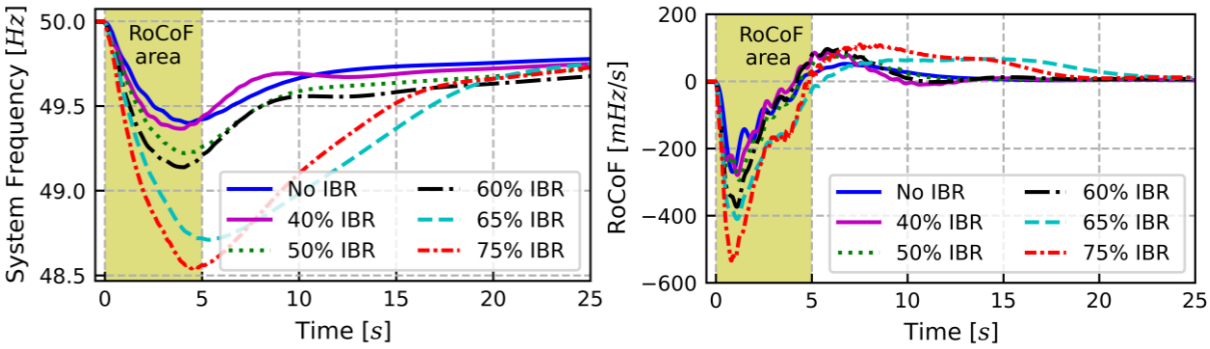


Fig. 4 Effect of inertia and POR reduction on the system frequency and RoCoF.

In fact, defining the level of inertia in a power system is, to some extent, subjective. This is because high or low inertia is characterized by a set of network conditions, including how the network is interconnected, the size of the system, largest infeed units, loss of generation/load, and percentage of IBRs. However, the conventional method for determining the power system inertia in real-time is to combine the inertia provided by each online synchronous machine. A quantified matrix is used in [25] to determine the current and future power system inertia levels, which presents many approaches currently employed by TSOs. This is because as part of the absolute change in system frequency, the TSO needs to manage the RoCoF variable.

2.3. RoCoF management

2.3.1. RoCoF in large synchronous areas

In 2013, the 'Future System Inertia' project was launched by the Nordic countries of the EU with the objectives: 1) develop a methodology to evaluate system inertia and frequency excursions; 2) develop a real-time inertia estimator; 3) harmonize inertia calculation; 4) investigate the impact of future generation and demand changes on system inertia [43]. Apart from determining the inertia constant for various power plant technologies, the study found that the inertia of the Nordic power system is projected to remain between 124 and 305 GWs, which is greater than the historical minimum inertia of 115 GW.s in 2009, indicating that serious RoCoF values are unlikely [44].

Following a typical generation loss, small synchronous areas would observe rapid and substantial frequency excursions, yet large synchronous areas would not see the same size of frequency deviations unless a significant disruption occurs, such as a system split. Currently, inertia levels are

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adequate for the normal interconnected operation of the EU power system while considering the normative loss of 3 GW [45]. In normal interconnected operations, RoCoF and frequency deviations are not a concern since the imbalances that occur are small in comparison to the present network inertia. In the EU, a loss of 3 GW credible disturbance would result in a 0.8 Hz frequency decline, considering the self-regulating effect of the load is 1%/Hz [46]. The network will not experience an extreme frequency drop unless a significant noncredible situation occurs, such as a system split during high power import/export conditions. However, an analysis showed that a reduction in the EU system inertia is noticeable which has lowered from 92% of all generations in 2012 to 79% in 2021 and is anticipated to decrease to 44% by 2030 [7].

Among the EU countries, Germany is a leader jurisdiction in the level of renewable power integration into its network, yet no challenges are observed. There are several reasons why Germany has been able to successfully integrate significant amounts of renewable energy, including the strength of its power networks, the flexibility of coal and nuclear plants, and well synchronous interconnection with neighboring countries [43]. This results in the overall nonsynchronous penetration remaining relatively low. Thus, the system is unlikely to exhibit issues related to RoCoF until a considerable high penetration of IBRs is integrated. However, to ensure dynamic stability in such circumstances, the mid-term solution may be to constrain cross-border flows between small and large synchronous regions [47]. Denmark is another example that regularly reaches or exceeds 100% nonsynchronous penetration of load, with synchronous condensers used to increase system rotational inertia. Due to the significant share of IBRs in the Danish system, new power plants larger than 1.5 MW must withstand RoCoF values of 2.5 Hz/s for 80 ms delay, which is a little increase over the previous 2 Hz/s limit [48].

Finally, due to IBRs, the North American Electric Reliability Corporation (NERC) and Hydro-Québec TransÉnergie (HQT) [49] observed a decrease in their respective system inertia and frequency response, which both proposed the use of synthetic inertia as a viable remedy. The Electric Reliability Council of Texas (ERCOT), in the United States of America (USA), has recently faced some technical issues due to the high IBRs [50]. Issues arose as overall ERCOT inertia declined from 152 GW.s to 130 GW.s between 2015 and 2017, which was partially attributed to the reduction in the net load [51]. This system is currently evaluating potential solutions to enhance frequency responsiveness, including renewables, as well as estimating prospects to secure it.

2.3.2. RoCoF in islanded power systems

In islanded power systems that are integrated with other large power systems via HVDC interconnectors, the impact of inertia reduction is especially significant. For example, the power system of Ireland, Great Britain (GB), and South Australia have already begun to have potential issues and are becoming more vulnerable to experiencing high frequency excursions and even partial blackouts during power outages [52]. In the Irish power and GB, RoCoF limits are determined by protection settings (i.e., loss of mains (LoM)) on the DG systems [41]. This protection system disconnects DG systems when the detected RoCoF breaches the predefined RoCoF and time delay thresholds.

In the GB, for example, the LoM was triggered in 2012 following a loss of generation associated with a high RoCoF [28]. This issue forced the TSO to re-dispatch the power system during times of high renewable generation. This required costly market intervention to replace nonsynchronous renewable sources with high inertia thermal power plants to ensure adequate level inertia and avoid excessive RoCoF oscillations. Another example of LoM activation was the cascading system failures in GB on August 9, 2019. The incident caused a considerable vector shift, which triggered the RoCoF protective relay and disconnected about 350 MW of DG systems from the network [34]. Currently, National Grid, the UK transmission system operator, is exploring measures to deal with declining

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inertia levels since it is anticipated to become a future concern. According to a National Grid analysis [53], the cost of operating the system to prevent LoM protection has increased from £60 million in 2018 to £150 million in 2019. Yet, it is unclear if National Grid would approach the issue by directly incentivizing inertia provision from synchronous sources in the future or through FFR services.

The problem of low inertia does not have a simple or single solution; rather, all possible options must be studied and weighed, whether now or in the future. Three alternative options are currently being considered by some TSOs [54]:

1) Limiting inertia floor during normal and transient conditions.
2) Modifying system equipment and grid codes to tolerate higher RoCoF values, as well as re-configuring the current RoCoF settings to new standards.
3) Incentivizing new forms of fast active power to preserve the LoM through the use of ancillary services.

The first option has been implemented in Ireland, which involves maintaining the required inertia level, keeping a number of online synchronous generators, and limiting the RoCoF by constraining the output of the largest generating unit. For instance, the TSOs of the all-island of Ireland presently maintain at least eight large synchronous sets online at all times [55]. As a minimum generation level exists for any dispatchable generator below which it cannot operate, this requires containing generation from renewable power to allow synchronous machines providing rotational inertia into the balancing mix. To account for this, in both jurisdictions, an operational metric known as SNSP was defined to ensure secure and prudent operation at any instant, as in (4) [35].

$$SNSP = \frac{(Non - synchronous\ generation + Net\ inerconnector\ imports)}{(Demand + Net\ interconnector\ export)}\ x\ 100 \qquad (4)$$

This is the maximum instantaneous penetration of nonsynchronous generators that can be allowed to operate at any time. The metric acts as a proxy for ensuring sufficient system services are available while avoiding a decline in system inertia below 23 GW.s [55]. The SNSP limit is currently set at 65%. However, to fulfill the 70% renewable electricity targets by 2030, an instantaneous SNSP of about 95% is expected [56]. The main challenge for GB and Ireland is curtailment owing to reduced inertia induced by the amount of wind generation. Renewable curtailment can occur when system operation rules, such as an SNSP limit, require TSOs to limit renewable output power generation.

The second option is now also being deployed by TSOs of some island power systems, as illustrated in Table 1. Historically, the setting in Ireland was -0.5 Hz/s over a rolling window of 500 ms [57], whilst in the GB, the standard was -0.125 Hz/s with no time delay [34]. This was in the context of systems with high inertia and many synchronous generators. However, with diminishing inertia, operating the system under the -0.125 Hz/s threshold has become increasingly difficult. This is because the amount of instantaneous "imbalance" power that will result in a RoCoF violation has been steadily lowering as the system weakens. Therefore, the RoCoF parameters in Ireland and GB have recently been relaxed to 1 Hz/s, both over a defining time of 500 ms to reduce the cost of ancillary services [53]. The new setting may temporarily alleviate the RoCoF security issue; nevertheless, it is not a silver bullet, as several countries have already begun planning for system conditions with RoCoFs above 1 Hz/s [28]. This is because a further reduction in the RoCoF setting may jeopardize the protection's ability to reliably identify LoM occurrences, rendering the protection's core function worthless.

Table 1. An overview of power system requirements to manage frequency stability.

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Regions	Contingency reserve	Inertia floor	|RoCoF| revised limits	SIR service	UFLS trigger
Ireland/Northern Ireland	500 MW	23 GW.s	0.5 -> 1 Hz/s	Yes	48.85 Hz
Great Britain	1250 MW	No – constraint to reduce largest loss is utilised	0.125 -> 1 Hz/s	No	48.8 Hz
USA ERCOT	2 largest generation ~2750 MW	100 GW.s (Online monitoring)	N/A	Yes	59.3 Hz
Australia*	Regional, SA = 350 MW	Regional limits under risk of islanding. SA = 6 GW.s TAS = 3.8 GW.s	Credible separation = 1 Hz/s Noncredible events = 3 Hz/s	No	47.6 Hz
Nordic	2 largest generators	120 GW.s (Online monitoring)	In islanding, the island must withstand 2 Hz/s -> 2.5 Hz/s	No	48.85 Hz

*SA: denotes South Australia, TAS: is Tasmania region. It is to be noted that while the National Electricity Market (NEM) does not currently have a system RoCoF limit, generators must comply with the National Electricity Rules (NER), which mandate RoCoF of 4 Hz/s for 250 ms for automatic access and 1 Hz/s for 1 s for minimum access standards [52].

Options one and two are only temporary solutions as other system needs (i.e., fault current, frequency and voltage regulation, grid restoration, etc.) will emerge while approaching extremely high IBR penetration [33]. Thus, either the third option or more cost-effective technique of lowering the RoCoF/increasing inertia, option one, must be examined. The most effective technique to deal with the instantaneous inertial response and actively arrest frequency deviation is to combine options one and three by adding SIR (i.e., synchronous condenser placement) and FFR services. Throughout the EU, only the TSOs of all-island of Ireland have developed, procured, and are now delivering joint SIR and FFR products. A study in [13] showed that curtailment issues can be partially addressed by reducing the inertial floor requirement (option one) and equipping renewable energy sources that can participate in the balancing market (option three).

Furthermore, Table 1 provides an overview of options one and two in some renewable dominant power systems to maintain frequency stability above the UFLS and LoM protection limits. As can be seen, frequency is the metric used in high inertia power systems to measure system security, and the key concern is to prevent the activation of UFLS. As a result, a certain amount of power is always available as a contingency reserve, which is sufficient to offset an unanticipated trip of a big generator or transmission line while still maintaining system balance. Nonetheless, RoCoF is a significant parameter in low inertia grids that use RoCoF relays on DGs to protect them against LoM.

3. Addressing power system operational challenges with high IBRs

In response to high renewable energy targets and the increasing vulnerability of decreasing system inertia, FFR services using various mitigation measures from demand response and IBRs have been developed in recent years.

3.1. International FFR services

Most of the power grids are currently facing lower inertia and more frequent losses than it has ever been previously. Moreover, future power systems are anticipated to be operated with substantial amounts of intermittent renewable resources, which could cause the frequency to deviate from nominal value more quickly and expose the system to high RoCoF exposure. Consequently, many

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jurisdictions contemplated revising frequency control frameworks to introduce new faster acting frequency response products as an option to deal with low system inertia and high RoCoF issues. Here technical details of some newly defined international FFR services are compared including the UK National Grid [58], Ireland EirGrid and Northern Ireland SONI [35], Australia AEMO [59], Canada HQT [60], and Alberta Electric System Operator (AESO) [61], Nordic [62], USA ERCOT [51] and PJM [63], and New Zealand Electricity Authority [64]. As shown in Table 2, there are currently no widely agreed upon FFR characteristics. The service is defined as having an activation time of no more than 500 ms following the time of the event and a full delivery time within 2 s [41].

National Grid has proposed four new fast response services that are more suited to fulfill future power system needs. First, dynamic regulation (DR) service requires tenders to provide a modulated frequency response activated within 2 s. The DR is a slow dynamic service that continuously regulates small variations of frequency, as demonstrated in Fig. 5 [58]. The service is not intended to arrest large frequency decline following contingencies but rather seeks to maintain continuous frequency regulation within a nominal value. This service is aimed predominantly at BESS assets to deliver a fast response service. Further, dynamic moderation (DM) is a fast dynamic service used to assist with a wider range of frequency regulation and keep it within operational limits. The requirements of these two services are symmetrical, which means the units need to provide the same amount of high frequency as low frequency response.

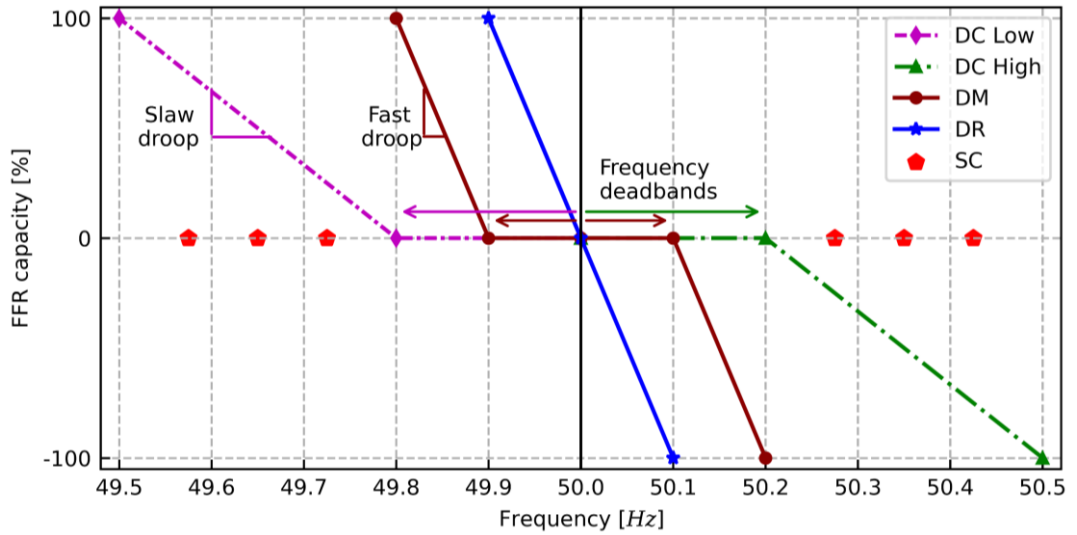


Fig. 5. Characteristics of dynamic and static FFR services in GB.

However, dynamic containment (DC) and static containment (SC) are designed to operate post-fault after a major frequency variation. In the DC service, the quantity requirements for low frequency and high frequency containments are different because the infeed and outfeed losses are usually different at any one time. This means the provision of DC does not need to be symmetrical, in fact, the provider could choose to participate in a low frequency or a high frequency alone. As shown, all the UK FFR services, except for SC, are droop type responses.

Droop-based FFR services are inspired by governor equipped synchronous generators' response characteristics, with the advantage that converters have smaller delays due to turbine dynamics, and hence, can respond more quickly than conventional power plants [36]. Droop control is a sort of distributed control that has the advantage of requiring no communication with other droop controlled units for coordination. Droop control enables each unit to contribute in proportion to its size, with the overall response being shared across providers. In other words, the output power of the FFR provider changes in proportion to the frequency deviation magnitude [65].

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Table 2. Fast frequency response service in different international system operators.

Power Systems	System operator	Service name	Service type	Deadband [±mHz]	Service trigger	Response time [s]	For duration [s]
UK	National Grid ESO	DC	Dynamic	200	Post-fault contingency	0.5	1200
		DR	Dynamic	15	Post-fault regulation	2	Infinite
		DM	Dynamic	100	Post-fault contingency	0.5	1200
		SC	Static/ multi-step	Various	Post-fault contingency	1	1800
Ireland and Northern Ireland	EirGrid and SONI	FFR	Dynamic	15-200	Post-fault contingency	0-2	10
			Static/multi-step	200			
Australia	AEMO	FRR	Dynamic	15	Post-fault contingency	0.1 - 1	6
			Static/five-step	200		1	N/A
Nordic	SOA	FFR	Static/single-step	300, 400, 500	Post-fault contingency	1.3, 1, 0.7	5-30
USA	ERCOT	FFR	Static	15	Post-fault contingency	0.25	900
		LR	Static	300	Post-fault contingency	0.5	3600
	PJM	FFR	RegD	N/A	Post-fault recovery	2	Sustained
Canada	HQT	FFR	Dynamic	100	Post-fault contingency	1.5	9
	AESO	FFR	Static	500	Post-fault contingency	0.2	N/A
New Zealand	Electricity Authority	FIR	Static	800	Post-fault contingency	1	60
		SIR		750	Post-fault recovery	60	900

DC, DR, and DM: are dynamic containment, regulation, and moderation. SC: is static containment. SOA: is a Nordic system operator agreement between Energinet, Fingrid, Statnett, Svenska kraftnät and Kraftnät Åland. LR: is a load regulation service similar to FFR. Both the LR and FFR in ERCOT have to restore within 3 hours and 15 minutes following the sustained duration, respectively.

The FFR services in the UK aim at both under and over frequency conditions, with the objective to address the most immediate requirement for fast acting response. This distinguishes it from the EirGrid FFR service, which is specially developed to limit the RoCoF and to defend under frequency post-fault frequency events, as shown in Table 2.2. The FFR service is defined in the DS3 program as an additional amount of power delivered or withdrawn from the network in response to response to frequency excursions to improve the frequency nadir and initial RoCoF [35]. One of the technical requirements for this service is that energy delivered between 2 and 10 s must be greater than energy drawn between 10 and 20 s after an incident [66]. Inverter-based resources such as wind, solar PV, batteries, and other emerging technologies, including demand response, appear to provide this service.

Nevertheless, the FFR service in Nordic countries is developed to manage the maximum instantaneous under frequency disturbances only as the highest concern is not the RoCoF. The Nordic static providers

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have three options for frequency activation levels and for total activation timeframes to participate in the service. It is to be noted that both the static Nordic service and the dynamic ERCOT appear to be more suitable for demand response technologies due to the required long duration to sustain post-fault.

The HQT, on the other hand, was the first grid operator to impose wind farms to be integrated with FFR features in 2010. It is specifically designed for wind farms with a capacity of more than 10 MW that needs to be operated in frequency sensitive mode (FSM). The system is deployed to be always available but only runs when a high frequency deviation occurs. Once activated, the generator boosts 6% more power than the nameplate capacity. Further, the AESO in Canada has recently developed another FFR service as part of a new technology pilot procurement competition to acquire any technology capable of rapid response. The service initially targeted load shedding and has now been expanded to incorporate BESS.

The TSO of New Zealand also currently procures two different products: Fast Instantaneous Reserve (FIR) and Sustained Instantaneous Reserve (SIR) from interruptible loads. This is similar to FFR service from interruptible loads, where the load must be curtailed within 1 s when the system frequency breaches 49.2 Hz and is required to sustain for at least 60 s following a disturbance. However, the SIR is designed to be delivered within a long time of 60 s following severe disturbances to restore frequency beyond 49.25 Hz. If the service is provided from the conventional spinning reserve, it must sustain for at least 15 minutes, while if interruptible loads provided the service, they must remain off until instructed by the TSO.

3.2. Frequency response service enabling technologies

Maintaining system frequency security and stability will require a fast response from a wider range of supplementary devices, including renewable technologies, demand response, and energy storage. Keeping the technical characteristics, as discussed in Section 4.1 in mind, the FFR service should be predominantly provided through demand response and IBRs. Traditional power plants (e.g., increasing synchronous capacity or adding new interconnectors) are not included in this discussion as they are well-understood. Instead, this section focuses on inverter-based technologies and other sources that may not have an inherent inertial response. The key properties of various resources are summarised in Table 3 [19], [39], [60].

As illustrated, currently, a response time within a range of 250 to 500 ms from the disturbance time is technically possible from a wide range of devices. The speed of response of these resources is a significant feature. This is because lower levels of system inertia result in a faster RoCoF (i.e., lower inertia means frequency drops quicker for a given circumstance). Thus, a fast response time is necessary to promptly arrest frequency drop before breaching the UFLS threshold limits. As the FFR service is typically performed via power converters, it can respond quickly, similar to the inertial response of synchronous generators [67]. The FFR service, unlike the inherent response of synchronous machines, is dependent on the active controls and can be tuned to the requirements of the power system within limits. However, the provision of FFR service by diverse technologies will be limited by the physics of the technology as well as their control characteristics.

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Table 3. Technologies with different frequency response service capabilities.

Technology	Power range (MW)	Energy rating (MWh)	Response time (ms)	Time of sustained power	Maturity
Wind turbines	Depends on the converter if overload and curtailment rates if curtailed	N/A	500 – 5000	Seconds (through virtual inertia), hours if operated below MPPT	Commercial but limited deployments
Solar PV	Depends on the converter if overload and curtailment rates if curtailed	N/A	100 – 200	Seconds to hours if operated below MPPT	Relatively immature for controlled active power applications
Flywheel (inverter)	0.1 – 400	Up to 5	≤ 4	Up to 15 min	Commercial but limited deployments
Lithium-ion Batteries	Up to 50 or more	Up to 20	Few to 40	Minutes to hours	Rapidly moving to mature
Flow batteries	~3, 50 possible	Up to 60	Up to 40	Seconds to hours	Relatively immature
Advanced lead-acid batteries	Up to 40	Up to 40	Few to 40	Seconds to hours	Based on the maturity of technologies
HVDC	Depending on converter capacity	Depending on the contracted period	50 – 500	Seconds to hours	Mature for triggering response functions
Super-capacitors	Up to 50	Up to 0.28	10 – 20	Milliseconds to hour	Relatively immature but developing quickly
Demand response	Depending on the contracted volume	N/A	40 – 60 Depends on load type	Seconds to hours (depends on the end user comfort)	Relatively mature technologies for tripping a large number of small loads

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3.2.1. Frequency support from wind power generation

Type 1 FSIG wind turbines have a limited inertial response during disturbances, which is inherently provided in response to the system frequency deviation similar to conventional generators. Commercially, a FSIG rated over 1 MW has typical inertia constant of 3 to 5 s, which is the time it takes for the generator to provide nominal output power from its stored kinetic energy [68]. However, unlike synchronous generators, VSWTs cannot inherently absorb or release kinetic energy stored in their rotors [69]. In addition, the generator cannot respond to frequency regulation signals to participate in system services. Instead, a supplementary control loop is required with the power controller to trigger inertial response and active power reserve during disturbances. The energy stored in the rotating mass can be managed via inertia control techniques to balance the grid power, and then the output of the generator is controlled in the same way as in synchronous generators [70]. Wind turbine inertial control can be further classified into two categories: inertia emulation, which exactly emulates the inertia response of synchronous generators, and deloading control technics, as shown in Fig. 6.

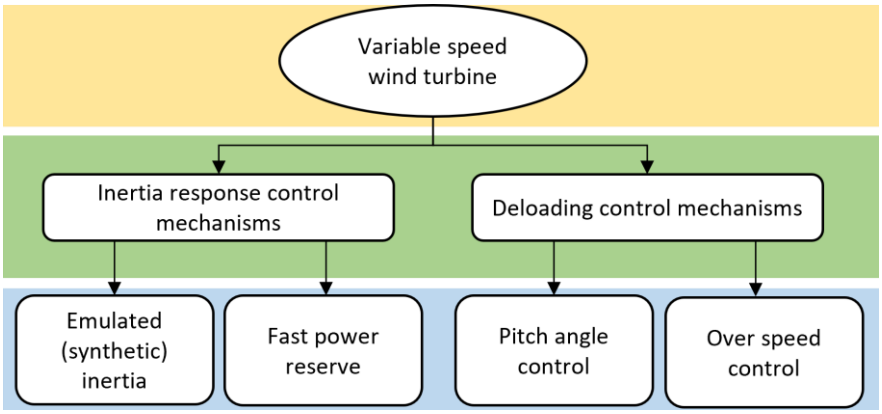


Fig. 6. Variable speed wind turbine frequency control technics.

a) Emulated Inertia: Inertia emulation introduces new control loops for releasing the kinetic energy stored within the rotating masses of the wind turbine. The power output of a wind turbine can be temporarily increased by utilizing the kinetic energy contained inside its blades to inject power temporarily. The boost, though, may not last for a long time. The emulated power provided by the wind turbine is relatively limited within a timeframe of 2 – 6 s [15] which is comparable to conventional SIR duration in the order of 2 – 10 s [71]. As the energy is provided from the rotor, the speed of rotation of the turbine reduces when the kinetic energy is exhausted. Slow turbine speed reduces aerodynamic lift, lowering supplied mechanical shaft torque and increasing electrical torque, potentially exacerbating the speed decline and resulting in blade aerodynamic stalling [72]. The inertia emulation period is followed by a recovery period during which the wind turbine experiences output power curtailment to re-accelerate the rotating speed of the turbine and avoid stalling.

Two possibilities are available for inertia emulations: one-loop, where the controller is activated based on RoCoF measurement, and dual-loops, where the additional power is proportional to both the RoCoF and the frequency divergence [25]. The schematic diagram of both controllers is shown in Fig. 7.

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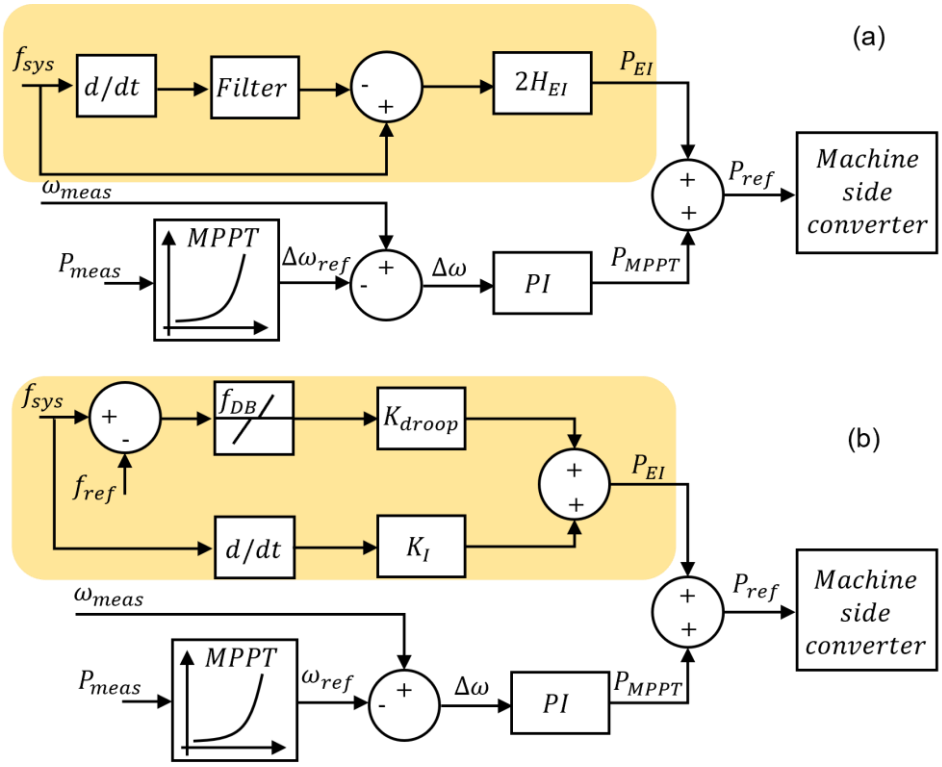


Fig. 7. Emulated inertia control methods (a) one-loop, (b) dual-loops.

The inertial power generated from the one-loop control technique can be mathematically represented using equation (5).

$$P_{EI} = 2H_{EI} f_{sys} \, {df_{sys}}/{dt} \tag{5}$$

where df_{sys}/dt is the system RoCoF in (Hz/s), H_{EI} depicts emulated inertia in (s) and f_{sys} is the system frequency in (Hz). The inertia control loop is enabled when the controller detects negative or positive RoCoF measurements during frequency deviations.

The dual-control loop, on the other hand, utilizes a combination of RoCoF and system frequency deviation (Δf_{sys}) control loops to provide additional power during contingencies. These controllers are mathematically represented as per (6).

$$P_{EI} = K_I \frac{df_{sys}}{dt} + \frac{1}{K_{droop}} \Delta f_{sys} \tag{6}$$

where K_I and K_{droop} are emulated inertia and droop control constant gains, respectively. In this case, the first term of (6) is used to give a short-term emulated power, similar to the one-loop control scheme. This is followed by the activation of the second term in (6), which serves as a droop response to restore system frequency to its nominal value. The latter controller requires the wind turbine to operate at a sub-optimal rate with a specific reserve during pre-disturbance. Notably, the VSWTs can exchange energy with the network much faster than synchronous generators due to the quick responsiveness of power electronic devices as well as the capability to operate within a wide range of controller gains. In light of this, a study in [69] deployed time varying control gains to alter the value of K_I and K_{droop} to regulate the amount of active power injected following frequency deviations. The larger gain values result in a better frequency deviation and RoCoF [73].

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b) Fast active power: Fast active power is utilized to limit the RoCoF and arrest frequency deviations. This method is also characterized as a fixed 10% of nominal active power for 10 s, regardless of wind speed [15]. The rapid power reserve, or short-term constant power, is generated from the kinetic energy stored in the rotating mass of the wind turbine. The active power reserve is obtained while the set point of the rotor speed is adjusted, as per (7).

$$P_{const,t} = {}^{1}/_{2}\, j\omega_{r,o}^{2} - {}^{1}/_{2}\, j\omega_{r,t}^{2} \qquad (7)$$

where $P_{const,t}$ blast active power injected, $t(t < t_{max})$ is the duration of the sustained power reserve since the frequency event began, $\omega_{r,o}$ and $\omega_{r,t}$ denote the initial and rotor speed at time t, respectively.

The operation of this controller begins when the system frequency deviation exceeds the predefined threshold limit. Then detecting and triggering scheme sends a control signal to bypass the MPPT and activate power shaping, as shown in Fig. 8 [74]. Once the disturbance is detected, additional power is injected; this is known as overproduction. When the kinetic energy release is accomplished, the speed recovery function restores the maximum power and returns the rotor speed to its pre-disturbance value. This process frequently results in the underproduction period, in which electricity is taken from the grid to restore the rotor speed to its target value. The nature of the emulated power and the characteristics of the energy recovery mainly depend on the prevailing wind speed during the time of the disturbance [72]. To avoid a second frequency dip produced by the decline in wind power to re-accelerate the turbine, other resources (i.e., BESS and demand response) may be necessary to coordinate with the response from the wind turbine [75], [76]. To avoid a sharp active power decline and without other means of support, the transition from overproduction to underproduction is required to be performed in a sloping way over a certain period.

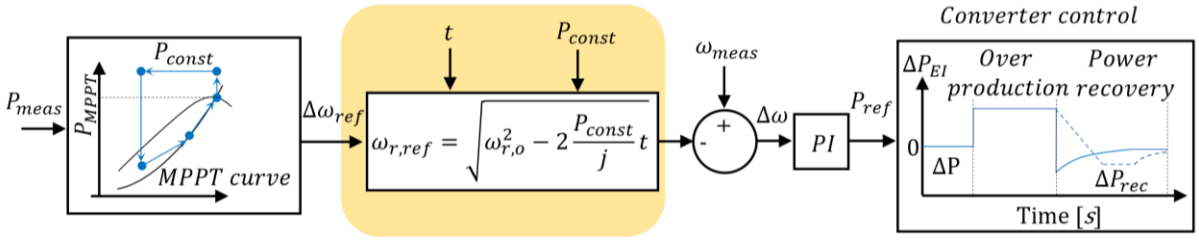


Fig. 8. Fast active power reserve control diagram and active power characteristics.

c) Deloading based droop response techniques: Alternative to inertia response, the wind turbine can be deloaded to reserve a certain headroom to participate in the system service markets [77]. This strategy, nevertheless, comes at a high energy cost. From the economic perspective, wind turbine technologies have been designed to operate at MPPT to extract the maximum possible energy. Consequently, there is no room for participation in the reserve market or support for dynamic frequency deviations. However, system requirements for enough reserve margin during contingencies force wind generators to operate below optimal level [78]. In the wind turbine deloading technique, a certain reserve is maintained by shifting the operating point of the turbine from the MPPT curve to a reduced power level. This is accomplished by either over-speeding the wind turbine or adjusting the pitch angle, as illustrated in Fig. 9.

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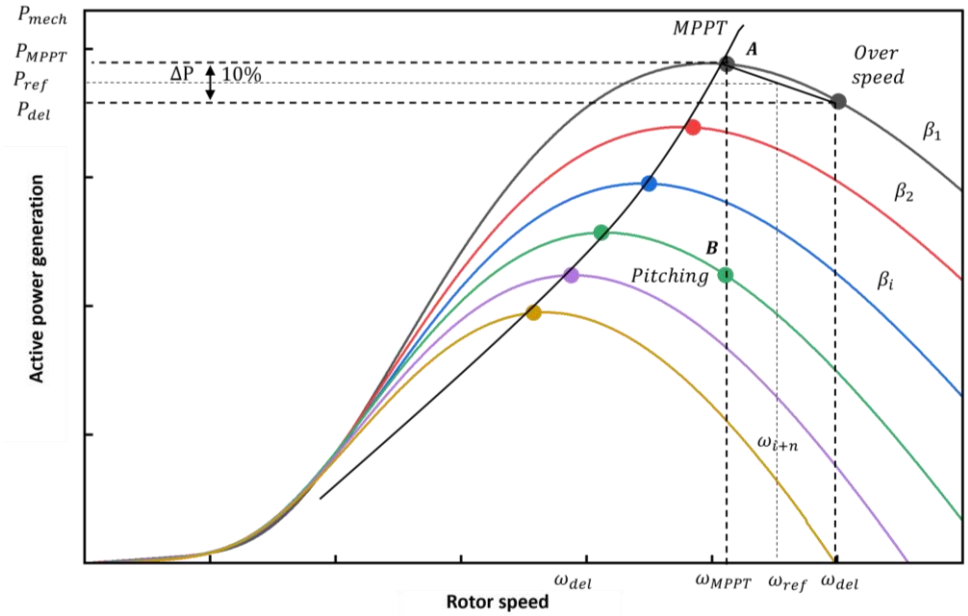


Fig. 9. Wind turbine deloading techniques include pitch control and overspeeding control.

In the speed control mechanism, the speed of the turbine is increased above the MPPT to regulate output power in response to system frequency under a fixed pitch angle [74]. This moves the provided power P_{del} to the right side of the maximum power P_{MPPT}. The rotor speed must be lowered from ω_{del} to ω_{MPPT} to release kinetic energy and to inject additional power ΔP. The wind turbine delivers a pre-set amount of active power, which is normally 10% [77], corresponding to frequency variation when the system frequency falls from the nominal value. The new operating point, using the reference power P_{ref} provided by (8), will be between β_1 and β_2.

$$P_{ref} = P_{del} + (P_{MPPT} - P_{del})\frac{\omega_{del} - \omega_{ref}}{\omega_{del} - \omega_{MPPT}} \tag{8}$$

Pitch angle control, on the other hand, regulates active power generation by adjusting the pitch angle of wind turbines while maintaining a constant rotor speed to compensate for frequency deviations [79]. This controller should ideally be turned on when the wind turbine generator reaches its rated speed and when the over-speed controller fails to do so. After that, the pitch angle controller starts to raise the wind turbine blades' pitch and moves the operational point from point A to a location between points A and B. The deloading control technique generally operates in two operational scenarios [80]. In a steady state, the VSWT operates at the optimal power curve by deriving the operating point from the look-up table. Nevertheless, the deloading mode will start up as soon as the deloading mode is switched on. The speed and pitch angle controllers will liaise in this situation to enable the wind turbine to store a certain amount of power in various modes.

The droop control mechanism can be used to manage the output power of the wind turbine in proportion to a system frequency deviation [81]. The droop control techniques considerably impact the frequency nadir and quasi-steady-state frequency. The active power generation is adjusted based on droop gain (K_{droop}) using (9). The droop technic is schematically shown from the upper part of Fig. 7(b).

$$\Delta P_{out} = \frac{1}{K_{droop}}(f_{sys} - f_{ref}) \tag{9}$$

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3.2.2. Frequency support from solar PV systems

Global solar PV capacity is expected to increase by approximately six times over the next ten years, from 480 GW in 2018 to 2840 GW by 2030, and to 8519 GW by 2050, or almost 18 times 2018 levels [82]. Asia, particularly China, would continue to dominate the PV market by 2050, accounting for more than half of all installed capacity, followed by North America at 20% and the EU at 10%. The majority of these are anticipated to be integrated into the distribution system, which will potentially reduce some of the conventional power plants currently used to control system frequency regulations. Solar PV uses MPPT techniques, similar to the VSWT, to enable the plant to run without any reserve power unless sophisticated control systems are built into the model. Despite the fact that PV plants are connected to a power system via power electronics converters, they can contribute to system frequency regulation through the following control techniques [24]:

1) Solar PV deloading (curtailing plant for reserve).
2) DC link capacitor response (virtual inertia response).

The most promising control method of frequency management for solar PV facilities is the deloading technique, which is accomplished by raising the PV voltage above the MPPT value. The PV array can curtail some reserve power by boosting the voltage from V_{MPPT} to voltage V_{del}, as illustrated in Fig. 10(a). This means that the PV plant will be operated at a deloaded operational point P_{del} which is below P_{MMPT}. In such an operational scenario, the curtailed power $\Delta P = P_{MMPT} - P_{del}$ will not be released until the system frequency deviates from the nominal value, as per Fig. 10(b).

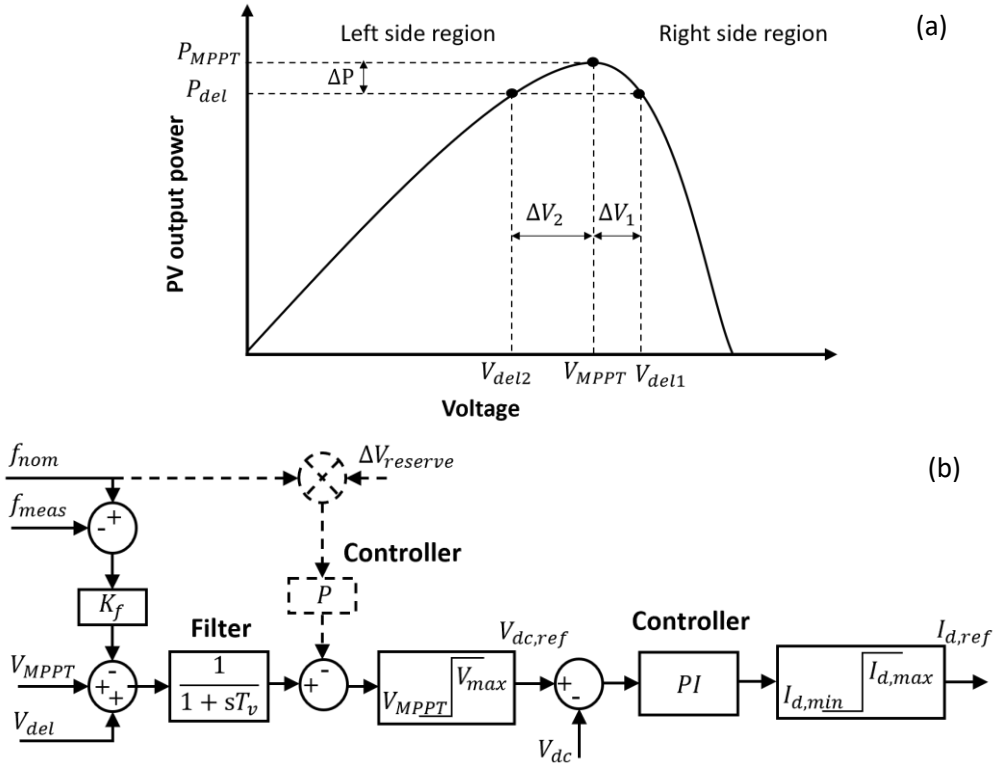


Fig. 10. Solar PV characteristics (a) deloading operational regions (b) active power control system.

Even though power reserve may be provided from both the right and left side operational zones, the right side region's de-loaded operation is preferred over the left side region's operation [83]. This is due to the converter's improved efficiency at higher voltage levels and the faster dynamic response brought on by the P-V curve's steeper slope. Furthermore, the left side region has a smaller power margin than the right side region since the converter's pulse width modulation (PWM) can only

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operate at a certain minimum voltage. However, due to potential instability challenges, de-loading control systems based on the regulation of the DC-link voltage typically employ the left side area to operate the PV generation [84]. In operational regions left side, MPPT, and the right side, the PV modules operate similar to a current source, a power source, and a voltage source, respectively.

The fact that the solid line controller in Fig. 10(b) does not account for the individual PV plant's available reserve is a significant weakness in its design. The fundamental drawback is that, regardless of the reserve power of each plant, all of them will deliver the same quantities of active power required for frequency stabilization. Consequently, some PV plants with low reserves reach the MPPT sooner and will not be able to contribute to frequency regulation further. To address this issue, the authors of [85] recommended a new improvement to the prior controller with the additional control signal reflecting the remaining reserve power $V_{reserve}$, as shown by the dashed lines. Moreover, [86] proposed an adaptive de-loading technique with three controller loops, including droop, active power-voltage matching, and vector. These extra loops make it possible to quickly regulate the frequency by modifying the PV system's output power.

3.2.3. High voltage direct current interconnectors

In comparison to traditional AC systems, HVDC transmission systems are known as an important driver for bulk power transmission across large distances with improved controllability. HVDC transmission systems can be classified as asynchronous interconnections, embedded transmission lines in synchronous systems, and offshore to onshore grid interconnections [87]. An asynchronous HVDC system is often used to link AC networks that operate at various frequencies or grids that operate at the same frequency but with different phase angles. The NORNED HVDC link between Norway and the Netherlands is an illustration of this kind of connection [88]. The ALEGrO link between Germany and Belgium is an example of an embedded HVDC system. HVDC synchronous interconnection is the connection between two locations in a synchronous area. HVDC links are increasingly recommended for transporting bulk power from offshore wind farms that are distantly located similar to the BorWin1 in Germany.

In a VSC-based HVDC converter, the active and reactive power can be controlled independently, which is a more recent development. This technology can restore AC electricity from a blacked-out state following a significant outage and support reactive power similar to STATCOM devices [89]. Furthermore, VSC converters are more dynamic, particularly with regard to power reversal, and the converter size is typically smaller due to the incorporation of smaller filters. For multi-terminal HVDC grids, VSC-based technology is seen to be a preferable option. To date, the majority of commissioned HVDC projects for offshore wind farms have deployed the VSC-based HVDC configuration [90]. The offshore converter station regulates AC voltage and frequency in islanded mode, and the converter can be used as a current source in the grid following mode [91]. Currently, the development of modular multilevel converter (MMC) leads the VSC topologies because of its high efficiency, superior output waveforms, and high scalability [92]. The HVDC integration strategy for offshore wind power now mostly uses MMC technology, which can be connected in grid forming (GFM) mode [93]. However, there are still certain issues with MMC technology, including as high power losses and costly switching equipment.

Similar to the frequency in the AC system, the DC voltage in the HVDC system reflects the active power balancing. Consequently, in a manner such as solar PV, the network's active power imbalances can be managed by adjusting the DC voltage of the station. Compared to the significant inertia in AC systems, DC systems' wires and capacitors only store an extremely small amount of energy. As a result, DC systems react to disturbances far more quickly, in the order of 20 ms, than AC systems [87]. In the ENTSO-E region, guidelines have also been developed for the use of HVDC systems to assist power system operation [94]. As an example, the all-Island of Ireland power system imports/exports power

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from the GB electricity grid via two HVDC interconnectors with a capacity of 500 MW each. The Moyle interconnector links Northern Ireland and GB, while the East-West Interconnector (EWIC) links the GB transmissions system to Ireland. Each interconnector provides a 75 MW reserve to protect against low and high-frequency response disturbances [55]. When the system frequency drops below 49.8 Hz, EWIC, for instance, acts with a rate of 500 MW/Hz up to maximum contracted volume at a frequency of 49.65 Hz [95]. Similarly, when the frequency reaches 50.2 Hz, EWIC responds with a rate of 500 MW/Hz up to the maximum agreed volume at 50.35 Hz.

3.2.4. Frequency support from asynchronous and synchronous flywheels

The flywheel is a mechanical device that stores kinetic energy in its spinning mass, which is proportional to the square of the rotational speed, as per (10).

$$E = {}^{1}/_{2}\, J\omega^{2} \qquad (10)$$

where J is the moment of inertia in (kg.m^2) and ω is the angular velocity in (rad/s). The flywheel uses grid power to spin a motor at high speeds and store energy, while to release energy, the flywheel momentum forces the motor to behave as a generator, decelerating the spinning down and injecting power into the grid proving inertia [96]. The amount of kinetic energy stored depends on the physical attributes and is proportional to the speed and inertia of rotating mass; therefore, they have inherent functions of inertia support.

An asynchronous flywheel storage capacity is independent of its age, and the technology is best suited to continuous charging and discharging processes. Thus they are used in the power system for short-term energy balancing such as frequency regulation to facilitate the integration of large-scale IBRs [97]. Short-term energy storage is commonly utilized to address short-term energy balancing and power quality issues that necessitate near-constant cycling to maintain a balanced power system. This application is continuous heavy cycling, which is well-suited for flywheels and some types of batteries. In comparison to the BESS, a flywheel can deliver a higher number of the complete depth of discharge cycles (approximately 200,000 vs 5000 for the BESS) and a higher energy density vs super-capacitors [98]. The key advantages of a flywheel are the fast-ramping capabilities, roundtrip efficiency, and lifespan built to last for 20 years or more. However, flywheels also possess disadvantages regarding energy storage, specifically the amount of energy they can store.

Synchronous condensers, on the other hand, are simply unloaded synchronous machines that operate in the motoring mode by default but may switch to the generating mode instantly if the frequency falls. This is an established technique that has been used for decades to provide reactive power support. However, advances in power electronics and semiconductor-based reactive power compensation, such as static var compensators, have largely dominated synchronous condensers as they provide reactive power compensation at a considerably lower cost and with better controllability [99]. All these have not affected the market of synchronous condenser deployments in many areas worldwide, where they have been treated as key to enabling renewable growth.

The market for synchronous condensers is expected to reach US$ 1.76 Billion in 2029 from its current value of US$ 1.01 Billion in 2022 [100]. This rise is mostly attributed to the incorporation of renewable IBRs as well as the growing use of synchronous condensers to support the reliable operation of power plants remaining from decommissioning process. Although IBRs are presently available to replace ancillary system services that would be lost as traditional power plants are decommissioned, there are a number of synchronous condenser functionalities that IBRs are unlikely to replace. Table 4 provides a comparison of the capabilities of both technologies [33], [42], [38].

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Table 4. Technical capabilities of synchronous condensers versus IBRs.

Synchronous condenser	Inverter based resources
Synchronous condensers have the voltage source characteristics, which means they can maintain the terminal voltage	Have physical limitations in terms of current and voltage control capabilities
Have a considerable overloading capability and short-circuit current injection	Only provide a small overloaded current rate for a short timeframe
Have inherent inertia support functions	Can support emulated inertia
Standard synchronous machine control mechanisms	The controller has a high bandwidth, can provide fast frequency and voltage regulation, and can perform numerous control schemes
They are physically limited and can only absorb a part of the full rated current	Can deliver the full rated current in both inductive and capacitive modes
Does not interact with other control systems	Issues with converter stability can arise from power converters' quick response time interacting with controllers of the same type and other networked devices

Furthermore, the redundant synchronous generators from the decommissioned power stations can also be converted into synchronous condensers with inertia augmentation through flywheel installations. This is because the decommissioning of mostly coal-based power stations has led to situations of low short circuit levels and reduced inertia, which has driven a resurgence in interest in synchronous condensers. The services provided by the synchronous condenser can, to some extent, make up for the loss of inertia and other services provided by the decommissioned power plants, ensuring that the system has enough strength to operate steadily. Synchronous condensers add inertia to the system by reacting quickly to power imbalances and contributing to RoCoF's constraints [101]. Synchronous condensers, in addition to inertia, can provide a variety of other services to tackle issues, including frequency control, transient stability (mitigating damping torque), supporting grid voltage, and short circuit strength [38]. They enable power system protection, reactive power support for dynamic voltage control with robust overloading and fault ride-through capability [41], and oscillation damping at the short circuit level [102]. These are the key reasons why this technology is being "resurrected" and attracting widespread interest, especially in islanded power systems (i.e., in Ireland, the United Kingdom, Australia, and Denmark), as shown in Table 5.

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Table 5. Recent synchronous condenser projects in islanded power systems.

Country	Project name	Commission date	Installed capacity	Objective	Reference
Ireland	Green Atlantic at Moneypoint by Siemens and Electricity Supply Board (ESB)	2022	400 MVA including large flywheel weight 130 tonne	Allow for higher penetration of wind power into the Irish grid	[103]
UK	Rassau Project in Wales by Siemens Energy and Quinbrook	2024	Combined 100 MVA synchronous condenser and 239 tonnes flywheel	In response to the rapid increase of renewable energy sources, the project is intended to play a significant role in National Grid's innovative approach to power system stability while attaining the Net Zero target	[104]
	Lister Driver Project in Liverpool by ABB and Statkraft	2021	Combined 67 MVAr synchronous condensers with 40 tonnes of a flywheel	Will provide more than 900 MWs of inertia and increase the instantaneously available inertia by 3.5 times	[105]
	Phoenix Project in Neilston by ABB project partner	2022	Combined 70 MVA voltage source converter (VSC) STATCOM with 70 MVA synchronous condenser	Referred to Hybrid-Synchronous Compensator (H-SC), can also provide reactive power support for black start	[106]
Australia	Davenport Project Manufactured by GE	2021	Two synchronous condensers rating 575 MVA/plant	Each plant provides 1100 MW.s and in total, they provide 4.4 GW.s of inertia. to maintain RoCoF below 3 Hz/s, secure operating envelope for voltage and transient stability	[107]
	Robertstown Project	2021	Two synchronous condensers rating 575 MVA/plant		
	EnergyConnect, interconnector by ANDRITZ	2024	Four synchronous condensers rating 120 MVA/plant	Inertia contribution is 7 MW.s/MVA	[108]
Denmark	Suðuroy, Faroe Islands Project by ABB	2022	Combined 7.5 MW BESS and 8 MVA synchronous condenser	Could enable 100% of the island's demand to be met by wind energy	[109]
	Sund, Faroe Islands	2023	synchronous condenser		

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It is worth noting that flywheels alone cannot alleviate frequency stability challenges but can be significant when combined with other mitigation measures such as static synchronous compensator (STATCOM) and BESS similar to the Phoenix project in Scotland. The features of the two systems can be combined to create a new technology that can do a variety of tasks and offer a range of services. This system would not contribute as much in terms of overloading capability as a standalone synchronous condenser with almost the same rating and coordinated control would be more challenging to handle.

3.2.5. Battery energy storage systems

It is commonly agreed that energy storage can contribute to grid stability if featured with pre-designed control requirements to participate in new services such as FFR and inertia response to make use of the accurate and fast response of converter interfaced storage devices. A grid-scale BESS can provide both frequency and voltage needs to the network following disturbances, as well as serve as an energy balancing resource. Its key advantages are the capability to work in four quadrants (i.e., consume or generate active and reactive power) and provide fast response times (usually between 100–150 ms or even less) during system contingencies, allowing it to deliver grid services with a high degree of flexibility [110]. For example, an 11 MW/5.5 MWh BESS developed in County Kerry, Ireland, can provide FFR and both active and dynamic reactive power services, with a response time of less than 150 ms [111]. Similarly, a 10 MW/5 MWh lithium-ion battery (LIB) was deployed by AES in 2016 at Kilroot Power Station in Northern Ireland, primarily as a prototype plan to react to system frequency disturbances, effectively providing both frequency regulation and smoothing [112]. This has been proved to provide ancillary services for grid stability during disturbances, as shown in Fig. 11, which corresponds to two real frequency events captured by phasor measurement units (PMU).

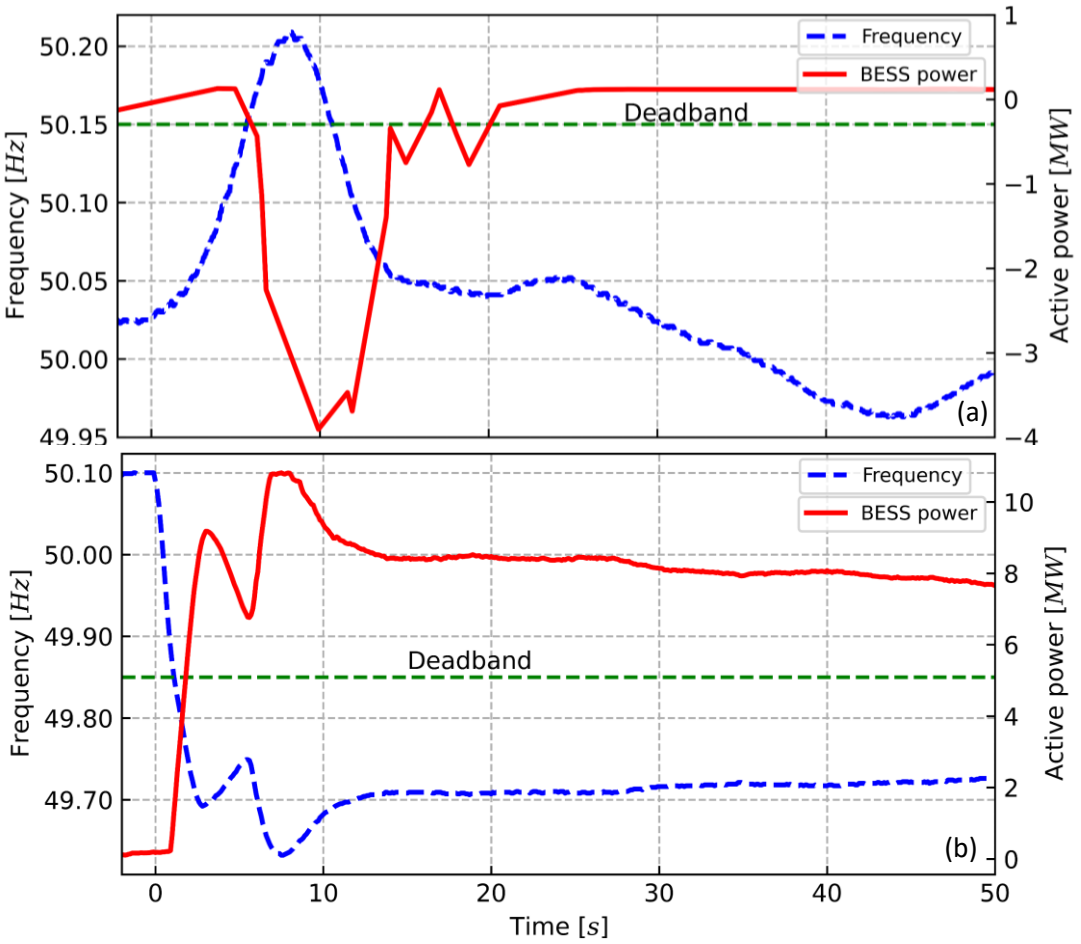


Fig. 11. Example of BESS active power response to frequency disturbances at Kilroot power station, (a): over-frequency occurred on 04/03/2016, (b): cascaded under-frequency occurred on 22/11/2021.

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As illustrated from the top panel, the BESS quickly responded to an actual over-frequency event that occurred in the Irish power system. As can be seen, the BESS started to absorb active power shortly when the system frequency breached the controller deadband limit of 50.15 Hz. The frequency controller remains consuming power from the network until the system frequency returns to below the controller sensitivity band. The bottom panel of Fig. 11 is the response of the same BESS site to a cascaded under-frequency event, roughly 6 s apart. As shown, during the first trip, system frequency dropped quickly, and the BESS did not respond until the deadband was reached, whereas the second trip occurred within the BESS deadband limit, resulting in a faster BESS response. Notably, due to the lack of system reserve, the frequency recovered to nominal after an extremely long time during this event.

BESS devices, and more broadly small-scale BESS (i.e., 10 kW), can reduce network restrictions and provide relief from imminent network improvements, which can negatively impact revenue streams and make it more difficult to determine appropriate installation locations [110]. The next stage of storage development begins with the deployment of hundreds of MWs on power systems at single sites. For instance, the Hornsdale Li-ion battery-based 100 MW/130 MWh power reserve in South Australia is well known in the world for effectively providing system services during low-inertia power dispatches, with 70% of its capacity reserved for providing FFR services to Australia's NEM, and the remaining capacity participating in the conventional energy market [113].

It is worth noting that IBRs (i.e., Solar PV, wind power, HVDC, and BESS) front-end converter systems can be operated using grid forming (GFM) or grid following (GFL) converter technologies. A GFM inverter, in its simplest form, is a converter that enables independent direct control of terminal voltage from grid voltage [42]. This characteristic sets GFM apart from standard GFL converters, where the current is the output that is directly regulated. In contrast to GFL technology, which needs a network signal for synchronization and current injection, GFM technology can generate a voltage signal even when there is no load or grid voltage. These features enhance the frequency response function based on the dynamic capability of the converter, as described in Table 6 [114], [38], [115], [116].

Table 6. Comparison between GFL and GFM grid support capability.

Grid following capability	**Grid forming capability**
Act as a current source and under typical operations, it is assumed that the grid has already been established.	Act as a voltage source while forming and maintaining a robust grid (e.g., contributing to system strength)
Require PLL to synchronize with the network	It can operate without PLL as well as with PLL to switch between different control modes
Follow network voltage to provide active and reactive power	Can provide the black-start capability from a complete shutdown
Control of current delivered to the grid using a direct-quadrature (*dq**) vector	Able to control voltage magnitude and frequency/phase using $P-f$, $Q-V$
Cannot actively operate at 100% IBR power penetration	Can hypothetically function with 100% IBR as well as cooperatively operate with grid following systems
Well established and globally standardized	Lacking operating experience from a systems operator point of view, as it is not standardized.
Commercially are not available with high overcurrent capability (even though technologically possible)	Commercially available to provide fault current of 200% of the rated value at the inverter terminals

**dq* control implies control concerning a three-phase synchronous reference frame. $P-f$ refers to active power and frequency. $Q-V$ reactive power and voltage.

Several GFM solutions have been deployed, including the use of virtual synchronous machines (VSM) [117], virtual oscillator based control (VOC) [118], power synchronization control [119],

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synchronverters, direct power control, and matching control [120], while the most basic control approaches involve allowing energy storage units to produce proportional p-f droop responses [42]. Although the GFM converters have been widely utilized in microgrids, standalone systems, and uninterruptable power supply (UPS) systems [121], their application has recently been extended to energy storage and IBRs [122]. For instance, the Dalrymple 30 MW/8 MWh BESS is the NEM's first grid-connected battery-based GFM converter, offering both regulated and competitive market services [123]. It is based on VSM technology, which strengthens the grid by mimicking the behavior and performance of synchronous generators, delivering both synthetic inertia and high fault current to allow more renewables to come into the network [124]. Finally, as summarised in Table 7, more pilot projects and demonstrators are being established around the world [125].

Table 7. Global pilot GFM projects.

Country	Project	Capacity	IBR type	Market type
UK	Dersalloch Siemens-Gamesa Project in Scotland	69 MW in VSM mode connected to 33 kV substation	Wind farm	Ancillary service, black start and islanded mode
Australia	Hornsdale Power Reserve, Jamestown	150MW/194MWh in VSM mode connected to 275 kV substation	Lithium-ion BESS	Ancillary service and energy market
	ESCRI South Australia Project by Hitachi ABB, near Dalrymple	30MW/8MWh in VSM mode connected to 132 kV substation	Lithium-ion BESS	Ancillary service and energy market
Switzerland	OSMOSE project, Ecole Polythecnique Fédérale de Lausanne	720 kW/560 kWh, connected to MV 0.3 kV substation	Lithium-Titanate Oxide BESS	Renewable integration
	OSMOSE project, EPFL Campus	25 kWh, connected to 400 V microgrid	Lithium-Titanate Oxide BESS	Renewable integration
France	OSMOSE project, RingoLab	6 x 0.5 MVA/60 min BESS and 6 x 1 MW/10 s supercapacitors	Hybrid energy storage system (HESS)	Renewable integration
Spain	La Plana Siemens-Gamesa Project	3 x 435 kW/145 kWh Lithium-ion BESS and 1 x 120 kW/400 kWh Redox-Flow	Hybrid	Ancillary service, black start and islanded mode
Portugal	DEMOCRAT Project	2 x 200 kW/109 kWh, connected to LV 400 V and MV 15 kV	Lithium-ion BESS	Black start, energy market and islanded mode
USA	SMA St. Eustatius Project	5.9 MWh	Lithium-ion BESS	Renewable integration and islanded mode
	NREL Campus Colorado Project	1.25 MW/1.25 MWh and 1 MW/1 MWh connected to 13.2 kV	BESS	Ancillary service and renewable integration

3.3. Frequency response activation techniques

The power system frequency response services are typically classified into two parts based on the activation method (e.g., droop and direct static step changes in the reference power), as shown in Fig. 12.

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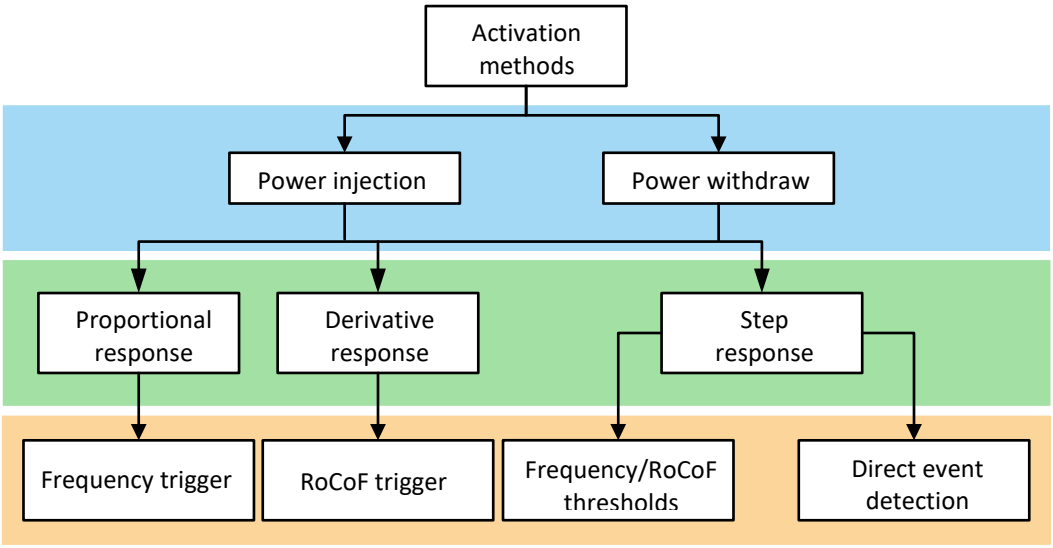


Fig. 12. Methods deployed to activate frequency response services.

The fundamental principle of droop control is to raise the active power reference when the grid frequency is lower than the nominal value and lower the active power reference when the grid frequency is higher than the nominal value [31]. This can have linear features, as illustrated in Fig. 13(a) [126]. The predetermined droop coefficient, frequency limit, power limit, and active power reference are used to regulate the active power reference. Droop parameters between 3-5% and a deadband of no more than 17-36 mHz are often applied [19].

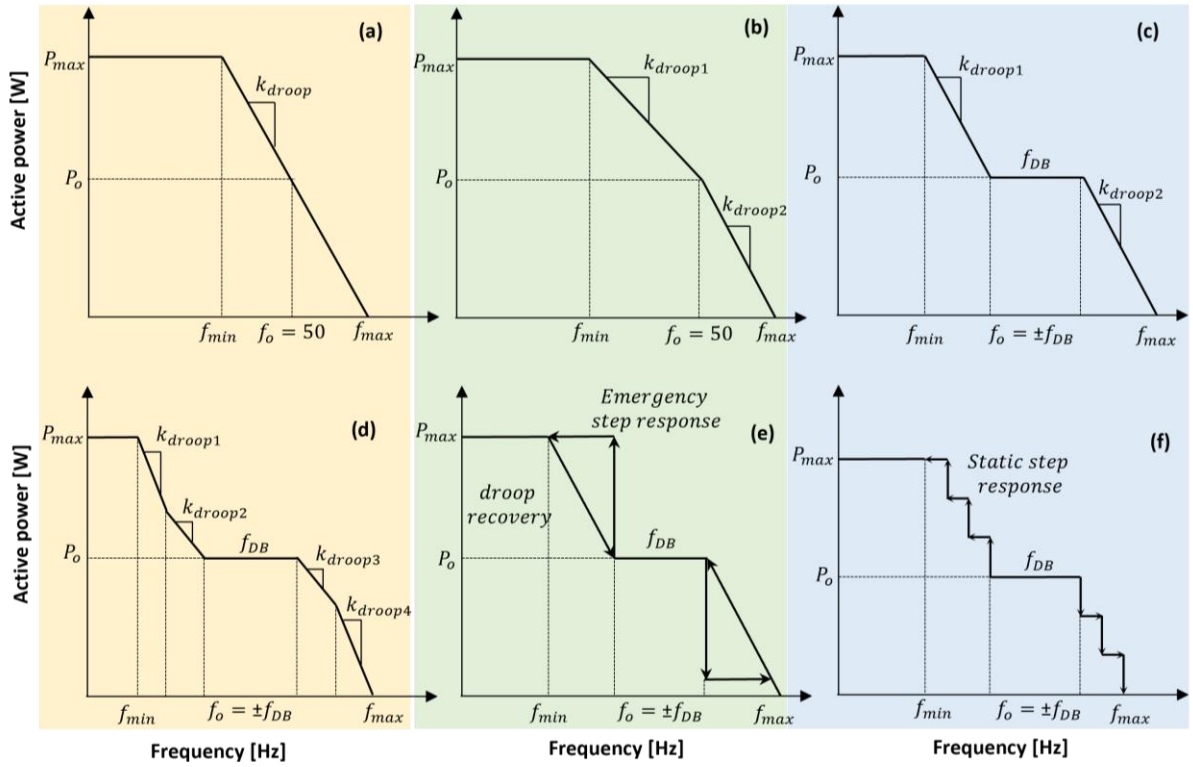


Fig. 13. Frequency droop control methods.

Pre-curtailment is necessary for all upward reserve services provided by IBRs powered by renewable energy, which results in lost energy sales. To alleviate the effect of this issue, the droop coefficients can be treated individually and differently in the over-frequency and under-frequency domains, as

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illustrated in Fig. 13(b). A deadband is often added to the *p-f* characteristics to avoid frequent activation caused by a little deviation in measurement tolerance or sensor noise, as shown in Fig. 13(c) [127]. Nevertheless, these droops adjust the active power at a fixed rate with respect to the frequency deviation. This can be further modified using the pointwise approach, in which the droop curve is created by connecting the dots, and can also be used to express the *p-f* linear characteristics [128]. Fig. 13(d) illustrates a segmented droop control method where a smaller droop coefficient is utilized to restore the little frequency deviation and a bigger droop coefficient is utilized to recover the extreme frequency deviations.

Nonetheless, this kind of droop may not be effective to provide FFR service as they are specially designed for slower response POR. From the perspective of TSO, an aggressive droop with a narrower deadband can boost FFR unit contribution and swiftly rebalance the grid frequency [129]. The frequency magnitude [130] or RoCoF [131] are commonly used to trigger services based on the static step response. The active power is suddenly injected into/curtailed from the system when the frequency or the RoCoF exceeds dead-band thresholds, as in Fig. 13(e). This kind of response can also be based on an emergency static step, in which the system operator delivers a centralized activation signal to a portion of demand causing it to trip in a single step. However, in order to avoid a second frequency disturbance following the first due to abrupt power recovery, the active power must be progressively restored to typical pre-contingency operation [16]. Engineers are always reluctant to change active power in a single step due to concerns about system oscillatory behavior. To mitigate this issue, the active power can be activated in multiple steps utilizing a variety of frequency threshold limitations, as shown in Fig. 13(f). In [132], a two-stage emergency control technique on RoCoF constraints was used to arrest frequency deviation.

When compared to droop controllers, this sort of control has the advantage of reacting to the specified quantity of power at the fastest rate once activated. Under low inertia dispatches, the quick power balancing due to the step response of some sources has a large potential to reduce spinning reserve requirements [133]. This means that the step response assists frequency recovery in a way that droop based response cannot [16]. However, because it can only reduce the power imbalance, this form of control can only assist with extreme frequency deviations. It cannot control the frequency to the desired level because the response is not continuously sensitive to the frequency deviation (as with droop, which has its own set of issues). To achieve the best performance, coordination among all participating resources may be essential [134]. Furthermore, as the static step response is a one-shot, it takes some time to reset and re-arm it once it has been activated. In contrast, as long as there is headroom and energy available, droop based control system will respond to the frequency measurement.

Furthermore, using frequency signals to activate demand response may be more accurate than utilizing RoCoF. This is because RoCoF is the derivative of the measured frequency and the second derivative of the phasor angle of an observable signal (e.g., voltage or current). This implies noisy measurement and cannot be performed instantaneously since high fidelity requires sample interval filtering [135], [136]. Although there are numerous techniques with varied accuracy and performance trade-offs (e.g., the interval between zero crossings and fast Fourier transforms), there are always concerns about false triggering based on RoCoF [25], [137]. Furthermore, as we incrementally lower the inertia of our power systems and thus problems with synchronous generators are exposed when there are no others to compensate them. In such circumstances, even minor disruptions can cause extreme nadir and rapid RoCoF [131]. During ultrahigh RoCoF levels caused by major disturbances, the FFR service triggered through system frequency measurement is unlikely to achieve the desired outcome [39]. This type of response may result in excessive wear and tear on over-frequency as a result of tripping demand greater than the required capacity.

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As a result, it is expected that FFR addressing low inertia conditions to be triggered mainly, if not entirely, by direct event detection rather than local frequency or RoCoF detection [39]. However, some service providers may continue to use frequency measurements to trigger to preserve system security. Direct event detection could be enabled with a communication response trigger or through wide-area broadcast options [138], [139]. Both strategies necessitate detecting the onset of the disturbance (e.g., a breaker opening on the AC lines or current zeroing on the interconnectors). The information is then transferred immediately to the FFR provider in the former control mode. In contrast, in the latter, data must be transmitted to transmission or distribution substations using high-speed communication before being disseminated to individual FFR providers. In both approaches, pre-programmed action is required to inject arresting energy into the grid. These methods bypass many of the problems with measuring frequency and/or RoCoF quickly and securely. However, they come with their own sets of challenges, such as relying on grid topologies, communication system performance, and security.

3.4. Distributed generation in IEEE standards

The typical view of electricity infrastructures is of grid-scale central power plants capable of generating one direction electricity flow through high-voltage transmission lines over long distances to distribution networks, which are then distributed to a large number of consumers. Due to the introduction and advancement of DG based IBRs, the traditional system structure has been altered, resulting in two-way power flow due to dispersed IBRs throughout the network. Consequently, several organizations have established technical standards for integrating IBR interconnections to ensure system stability, as depicted in Fig. 14. For example, the IEEE Standards 1547 series has served as a fundamental document for integrating IBRs into the power system. The first series IEEE 1547-2003 primarily specifies the minimum technical requirements, such as power quality, frequency deviations, and voltage levels as well as safety and protection, that are universally required to ensure secure and reliable connectivity [213].

Due to the potential impact of the increased penetration of IBRs on the power system operation and dynamics, standard 2013 presents an overview of distribution secondary network system design, discusses considerations for connecting IBRs to networks, and suggests potential solutions for IBR connectivity in network distribution systems [214]. The objective of standard 2014 was to manage the potential impact of IBR operation at the system level. Nonetheless, this project was withdrawn later in 2016. The interconnection standards in place were adequate for small amounts of power flows from IBRs but not for the large number of IBRs anticipated to support the fundamental change within the distribution system. IEEE standards and grid codes worldwide have been fundamentally changing to require IBR to provide power system support services and remain connected for a particular time when a grid malfunction occurs to boost the hosting capacity of renewable IBRs.

This has resulted in a series of amendments (i.e., IEEE 1547a-2014 and IEEE 1547.1a-2015) to support low/high voltage/frequency ride-through functions (L/HVRT and L/HFRT) in response to faults at the terminal of IBR units [215], [216]. To handle compatibility and associated interfaces between IBR and the network, standard IEEE 1547a-2014 demands the power system support functions, including frequency-droop regulation and inertial response to the RoCoF. In IEEE 1547-2018, if reserve power is available, the IBR must provide frequency-Watt ($P-f$) control, but it is not required to maintain upward or downward reserve to enable $P-f$ [217].

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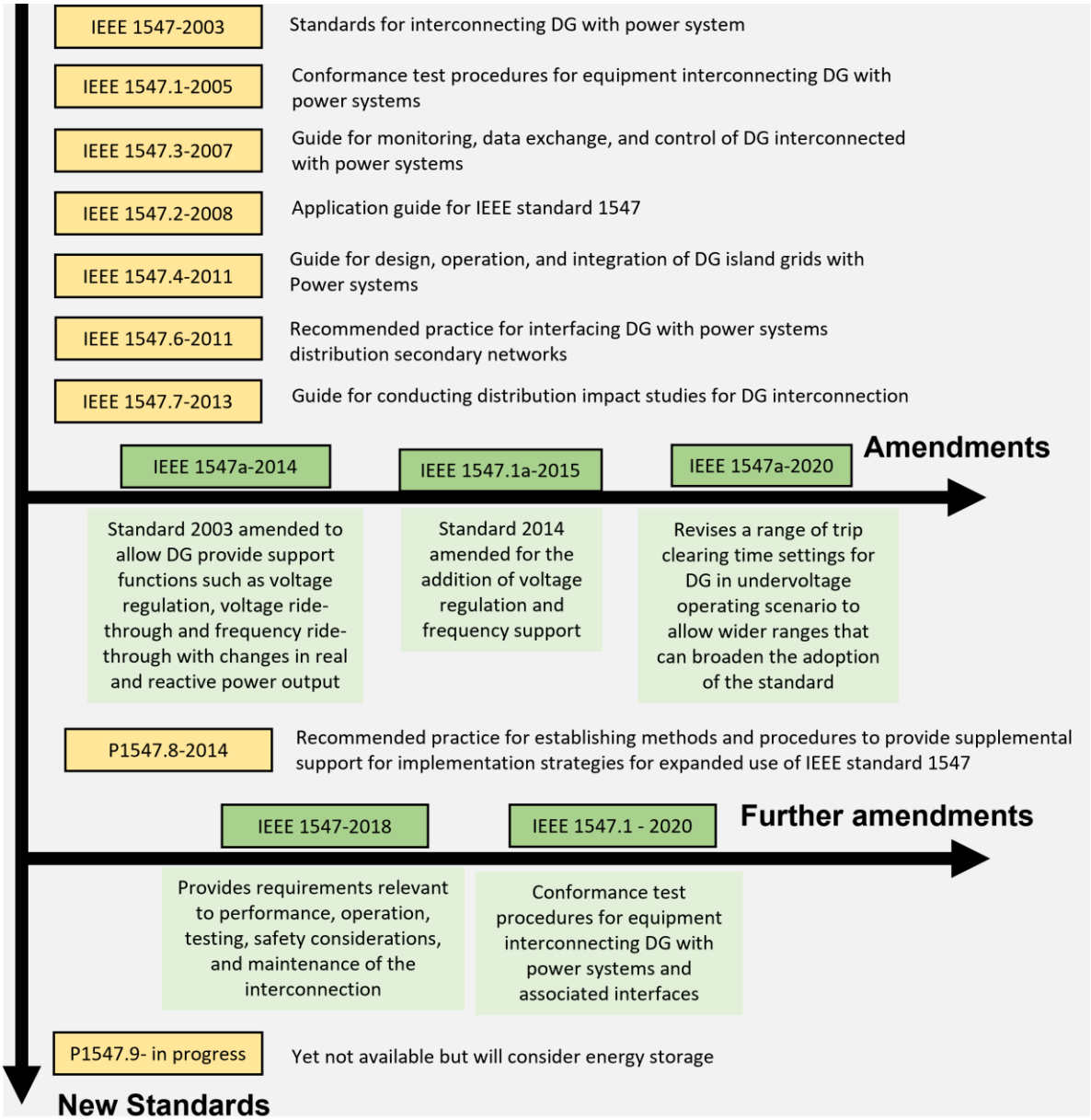


Fig. 14. The IEEE standard developments for managing distributed generation operations.

Furthermore, IEEE 2030 is a foundational set of system-of-systems standards developed from scratch in 2011 to govern network connectivity and interoperability [218]. It has been modified to allow for the continued implementation of modern communications and information technologies that provide interoperability solutions for IBR connectivity as well as smart end user appliances. Detailed guidance is provided in series 2030-1 up to 2030.3 for power system support functions by electric transportation infrastructure and interoperability of energy storage systems. The revisions primarily concern the ability of two or more systems to exchange and use information securely and effectively. These standards have addressed the participation of IBR technologies including the potential grid-scale data center in the future power system operation with high penetration of renewable IBRs. Finally, a clear path has yet to be identified for IBR equipped with different converter technologies (i.e., GFL and GFM), including their share and allocations to securely operate renewable rich power systems.

4. Discussions and recommendations

With the ongoing decline in system inertia, maintaining the frequency deviations and RoCoF, in an effective manner, become increasingly difficult. System inertia and frequency control mechanisms are

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among the major concerns facing future power system operations. A progressive drop in system inertia results in significant frequency deviations with extreme RoCoF during power imbalance such as the loss of an interconnector, a large load, or a generator trip. Furthermore, the traditional POR service is becoming inadequate, and providing this service from the spinning reserve of synchronous generators may result in a considerable increase in operational costs in the future. Given the growing use of renewables worldwide, the value of capacity and ancillary services will rise, as will the potential for IBRs to participate in the ancillary service markets. Grid operators are expected to work toward mandated services, and grid codes, to ensure that future power system needs can be met. This is because the source of services to meet these needs will change from conventional power plants to IBR, synchronous condensers, and energy storage devices, as summarized below:

- One of the widely discussed solutions is to curtail variable wind and solar PV power plants during low demand times and modify reserve policy and extra static reserve at times of low system inertia and high renewable generation. These technologies are well placed to provide a variety of system services required for alleviating relevant network scarcities. This is crucial since these mitigation options will be available at times when renewable energy production is high, and the scarcities are often worse because traditional service providers like conventional synchronous units have been replaced. However, maximum power generation will be required from renewable sources in future power systems [140]. This means a more radical solution needs to be emerged with special control mechanisms to inject/withdraw power in response to system frequency variations.

- New flexibility sources to accelerate the transition to the deployment of high IBR via participation in FFR service markets are required. These new sources will come from new technologies like utility-scale BESS equipped with both GFL and GFM converters. GFM devices are conceptually intriguing because of their ability to address various limitations of GFL converters. The voltage source characteristics of the GFM converter reveal that they can boost system strength and some additional fault current. However, this area still requires extensive analysis related to their practical implementation. Among them, are the transition between islanded mode and grid-connected mode, identifying the right share and location contemplating grid topology, strength, inertia changes, and development of regulations [130].

- While a range of various technologies and techniques can be used to address particular scarcities, specific technologies can be used to address a variety of technical concerns. As an example, synchronous condensers, in addition to voltage support, can aid to postpone frequency nadir, allowing slower-acting reserves to be activated prior to its occurrence. Their advantages are magnified when utilized in conjunction with other mitigation techniques. Thus, the appropriate mix of solutions must be thoroughly assessed for the system, while considering the synergy between various mitigation technologies.

5. Conclusions

This paper provided an extensive overview of power system stability issues as synchronous power plants are rapidly being displaced by IBRs. This research evaluated a detailed power system dynamic frequency stability issue associated with integrating IBRs and solutions in prospective technological pathways to assist system operators. With more widely acknowledged IBR values and benefits (i.e., improved controllability and grid resilience) global power systems are transitioning to a new energy mix that relies less on fossil fuels and nuclear while incorporating more energy storage, demand response, and IBRs. From reviewing the key FFR enabling technologies, it was possible to conclude that power systems require new sources of flexibility to accelerate the transition to high IBRs via participation in FFR service markets. These flexibilities can come from IBRs with advanced converter control mechanisms or from synchronous condensers. Finally, addressing power system scarcities will necessitate a holistic approach based on system-level thinking while introducing new markets

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incentivizing investment in technology that can help the electricity system transition to a cleaner and greener future.

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